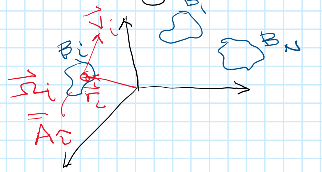
Full Body Problem

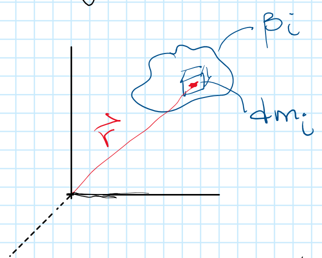
Sunday, February 15, 2026 6:46 PM

The gravitational FBP deals w/ N-bodies w/ mass that have finite densities, & can be considered to be rigid bodies. Include contact dynamics



Replace pt masses w/ finite density distributions.

Instead of specific $m_i \Rightarrow dm_i$ & $\mathcal{B}_i$ = mass distributions



$dm_i = \sigma_i(\vec{r})\,dV$

$$\sigma(\vec{r}) = \begin{cases} 0 & \vec{r} \notin \mathcal{B}_i \\ \sigma_i(\vec{r}) \neq 0 & \vec{r} \in \mathcal{B}_i \end{cases}$$

for a pt mass we have

$dm = m_i$

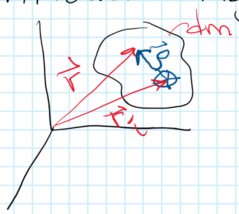
$\sigma_i(\vec{r}) = \delta(\vec{r}_i)$

$\mathrm{Vol}(\mathcal{B}_i) = \int_{\mathcal{B}_i} dV$; $\quad m_i = \int_{\mathcal{B}_i} \sigma_i(\vec{r})\,dV$

$$M = \int_{\mathcal{B}} \underbrace{\sigma(\vec{r})}_{\text{density}}\,dV = \sum_{i=1}^{N} \int_{\mathcal{B}_i} \sigma_i(\vec{r})\,dV = \sum_{i=1}^{N} m_i \quad \text{where } \mathcal{B} = \{\mathcal{B}_i \,;\, i = 1, 2, \ldots N\}$$

How do we define the location & state of each/all bodies?

Introduce rigid body orientation parameters along w/ positions



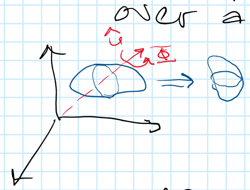
$$\vec{r}_i = \frac{1}{m_i} \int_{\mathcal{B}_i} \vec{r}\, \sigma_i(\vec{r})\,dV$$

$$\int_{\mathcal{B}_i} \vec{\rho}\, \sigma(\vec{r})\,dV = \vec{0}$$

Attitude Specification

To orient rigid body recall Euler's Thm

The orientation of any rigid body, relative to a given frame can be reduced to pure rotation over a fixed axis.



$\Rightarrow$ Axis-angle coord

$(\Phi, \hat{u}) \equiv$ Attitude DoF

$\uparrow$ rotation axis

- can use Euler angles but can have singularities

Directly related to quaternions $\bar{q} = \begin{bmatrix} \sin\frac{\Phi}{2}\hat{u} \\ \hline \cos\frac{\Phi}{2}\hat{u} \end{bmatrix}_{4\times 1}$ } 3 DoF for attitude of the body

Rotation dyads

$$\boxed{\bar{\bar{A}} = \cos\Phi\, \bar{\bar{U}} + (1-\cos\Phi)\, \hat{u}\hat{u} + \sin\Phi\, \tilde{\hat{u}}}$$

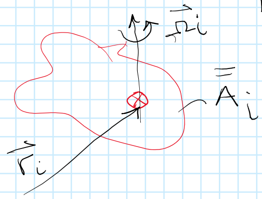
$$\boxed{\text{Rotation dyads} \quad \bar{\bar{A}} = \cos\Phi\, \bar{\bar{U}} + (1-\cos\Phi)\,\hat{u}\hat{u} + \sin\Phi\,\tilde{\hat{u}}}$$

body

$$\bar{\bar{A}}^T \bar{\bar{A}} = \bar{\bar{U}}$$

Time Rate of Change of Attitude

use the angular velocity of rigid body relative to inertial space

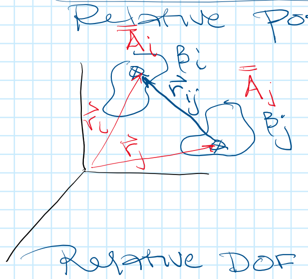


Time rate of change of $\bar{\bar{A}}_i$

$$\dot{\bar{\bar{A}}}_i = \bar{\bar{A}}_i \times \vec{\Omega}_i = A_i \cdot \tilde{\Omega}_i$$

$$\boxed{\vec{v}_i = \dot{\vec{r}}_i}$$

Relative Positions & relative orientations of NBP



DOFs

i: 3 pos + 3 attitude = 6 DOF

j: 3 pos + 3 attitude = 6 DOF

12 DOF

Relative DOF between 2 bodies

$$\vec{r}_{ij} = \vec{r}_j - \vec{r}_i$$

$$\bar{\bar{A}}_{ij} = \bar{\bar{A}}_i^T \bar{\bar{A}}_j$$

$\bar{\bar{A}}_j$ = goes from body $j \rightarrow$ inertial space

$\bar{\bar{A}}_i$ = goes from inertial space $\rightarrow$ body i

Inertial Integrals

0^{th}: $m_i = \int_{\beta_i} \sigma_i \, dV$ $\qquad M = \int_\beta \sigma \, dV$ $\qquad \sigma$ = density

1^{st}: $\vec{r}_i = \frac{1}{m_i}\int_{\beta_i} \vec{r}\,\sigma_i \, dV$ $\qquad \vec{R}_c = \frac{1}{M}\int_\beta \vec{r}\,\sigma\, dV = \vec{0}$

2^{nd} moments:

$$\bar{\bar{I}}_i^{cm} = -\int_{\beta_i} \underbrace{\tilde{\rho}\cdot\tilde{\rho}}_{\text{relative position vector}}\, \sigma_i \, dV = \int_\beta \left[\rho^2\, \bar{\bar{U}} \cdot \vec{\rho}\,\vec{\rho}\,\right] \sigma_i \, dV =$$

Inertia dyad in Body frame

$$\bar{\bar{I}}_i^{\mathcal{N}} = \bar{\bar{A}}_i \cdot \bar{\bar{I}}_i \cdot \bar{\bar{A}}_i^T =$$

Inertia in an inertial frame

Total Inertia rel. to system barycenter

$$\bar{\bar{I}}_i^{I} = m_i \tilde{r}_i \cdot \tilde{r}_i + \bar{\bar{A}}_i\, \bar{\bar{I}}_i^{cm} \cdot \bar{\bar{A}}_i^T$$

(assume $\vec{r}_i$ in inertial frame)

$$\boxed{\bar{\bar{I}}_i^{I} = m_i\left[r_i^2\, \bar{\bar{U}} - \vec{r}_i\vec{r}_i\right] + \bar{\bar{A}}_i\, \bar{\bar{I}}_i^{cm}\, \bar{\bar{A}}_i^T}$$

System Inertia

$$\bar{\bar{I}}^{I} = \sum_{i=1}^{N} \bar{\bar{I}}_i^{I} = \sum_{i=1}^{N}\left\{m_i\left[r_i^2\,\bar{\bar{U}} - \vec{r}_i\vec{r}_i\right] + \bar{\bar{A}}_i \cdot \bar{\bar{I}}_{cm} \cdot \bar{\bar{A}}_i^T\right\}$$

$$\boxed{\bar{\bar{I}}^{I} = \frac{1}{M}\sum_{1 \le i \le j \le N} m_i m_j \left[r_{ij}^2\, \bar{\bar{U}} \cdot \vec{r}_{ij} \cdot \vec{r}_{ij}\right] + \sum_{i=1}^{N} \bar{\bar{A}}_i \cdot \bar{\bar{I}}^{cm}\, \bar{\bar{A}}^T}$$

$$\boxed{\bar{\bar{I}}^{I} = \frac{1}{M}\sum_{1\le i<j\le N} m_i m_j \left[r_{ij}^2 \bar{\bar{U}} - \vec{r}_{ij}\vec{r}_{ij} \right] + \sum_{i=1}^{N} \bar{\bar{A}}_i \cdot \bar{\bar{I}}^{cm} \bar{\bar{A}}^T}$$

Polar moment of Inertia

For general Inertia $\quad I_p = \frac{1}{2}\mathrm{Tr}[\bar{\bar{I}}] \Rightarrow$ apply to $\bar{\bar{I}}^{I}$

$$I_p = \frac{1}{M}\sum_{1\le i<j\le N} m_i m_j r_{ij}^2 + \frac{1}{2}\sum_{i=1}^{N} \mathrm{Tr}[\bar{\bar{I}}^{cm}]$$

$\Rightarrow$ Lose all rotational DOF for the polar moments.

Moments of Inertia relative to the angular momentum

total Ang momentum of the system

$$\vec{H} = \sum_{i=1}^{N} m_i \vec{r}_i \times \vec{v}_i + \sum_{i=1}^{N} \bar{\bar{A}}_i \cdot I_i^{cm} \cdot \Omega_i = \text{constant} = H\hat{H}$$

$$\boxed{I_H = \hat{H}\cdot\bar{\bar{I}}^{I}\cdot\hat{H} = \frac{1}{M}\sum_{1\le i<j\le N} m_i m_j \left[r_{ij}^2 - (\vec{r}_{ij}\cdot\hat{H})^2 \right] + \sum_{i=1}^{N} \hat{H}\cdot\bar{\bar{A}}_i\cdot\bar{\bar{I}}^{cm}\cdot\bar{\bar{A}}_i^T\cdot\hat{H}}$$

inertial about ang momentum axis

Amended Potentials

$$\boxed{\mathcal{E}_p = \frac{H^2}{2I_p} + U(\vec{r}_{ij}, \bar{\bar{A}}_{ij})}$$

Note $I_p(r_{ij})$ only func of relative positions

$$\boxed{\mathcal{E}_H = \frac{H^2}{2I_H} + U(\vec{r}_{ij}, \bar{\bar{A}}_{ij})}$$

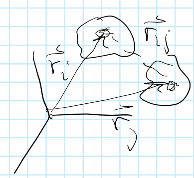
$I_H(r_{ij}, \bar{\bar{A}}_{ij})$

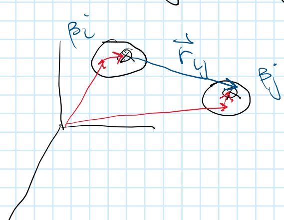
Potential Energy

NBP $\quad U = G\sum_{1\le i<j\le N} \frac{m_i m_j}{r_{ij}} = -\frac{G}{2}\sum_{i=1}^{N-1}\sum_{\substack{j=1\\ j\ne i}}^{N} \frac{m_i m_j}{r_{ij}}$

Let $\quad m_i \to dm_i = \sigma_i(\vec{r})\,dV$

$$\vec{r}_{ij} = (r_j^{com} + \rho_j) - (r_i^{com} + \rho_i)$$





$$U = -\frac{G}{2}\sum_{i=1}^{N-1}\sum_{\substack{j=1\\ j\ne i}}^{N} \frac{m_i m_j}{r_{ij}}$$

$$\sim -\frac{G}{2}\int_{B_i}\int_{B_j} \frac{dm_i\,dm_j}{\vec{r}_{ij}}$$

what happened to $j \ne i$?

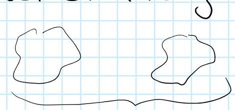
Choose inertial frame for computation

$$\vec{r}_{ij} = \vec{r}_{ij}^{\,com} + \vec{\rho}_j \vec{\rho}_i$$

$$\boxed{\vec{r}_{ij} = r_{ij}^{com} + \bar{\bar{A}}_j\vec{\rho}_j + \bar{\bar{A}}_i\vec{\rho}_i}$$

Break the integration over each body separately

$$U = -\frac{G}{2}\int_{B'}\int_{B''} \frac{dm'\,dm''}{r_{ij}}$$



each integral acts integrated over both bodies

$B = \{B_i, B_j\}$

$$U = -G\left[\int\int \frac{dm_1'\,dm_1''}{} + \int\int \frac{dm_2'\,dm_2''}{} + \int\int \frac{dm'\,dm''}{} \quad \int\int \frac{dm''\,dm'}{}\right]$$

integrated over both bodies

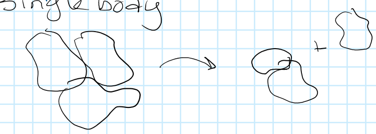
$$\mathcal{U} = -\frac{G}{2}\left[\int_{\beta_1'}\int_{\beta_1''}\frac{dm_1'\,dm_1''}{r_{11}} + \iint_{\beta_2'\beta_2''}\frac{dm_2'\,dm_2''}{r_{22}} + \iint_{\beta_1'\beta_2''}\frac{dm_1'\,dm_2''}{r_{12}} + \iint_{\beta_2'\beta_1''}\frac{dm_1''\,dm_2'}{r_{21}}\right]$$

New potential energy terms
"Self potentials"

represent the PE held in a finite mass distr.

$$\boxed{\mathcal{U}_{ii} = -\frac{G}{2}\iint_{\beta_i\beta_i''}\frac{dm_i\,dm_i'}{|\vec{\rho}_i - \vec{\rho}_i'|}}$$

single body



Note:

$$-\frac{G}{2}\int_{\beta_1}\int_{\beta_2}\frac{dm_1\,dm_2'}{|\vec{r}_{12} + \bar{\bar{A}}_2\vec{\rho}_2 - \bar{\bar{A}}_1\vec{\rho}_1|} = -\frac{G}{2}\iint_{\beta_2\beta_1}\frac{dm_2\,dm_1'}{|\vec{r}_{21} + \bar{\bar{A}}_1\vec{\rho}_1 - \bar{\bar{A}}_2\vec{\rho}_2|}$$

Leads to the mutual potential over all bodies

$$\mathcal{U}_{ij} = -G\int_{\beta_i}\int_{\beta_j}\frac{dm_i\,dm_j}{|\vec{r}_{ij} + \bar{\bar{A}}_j\vec{\rho}_j - \bar{\bar{A}}_i\vec{\rho}_i|} \qquad \mathcal{U}_{ij} = \mathcal{U}_{ji}$$

$$\mathcal{U}_{ij}(\vec{r}_{ij}, \bar{\bar{A}}_j, \bar{\bar{A}}_i) = \mathcal{U}_{ij}(\vec{r}_{ij}, \underbrace{\bar{\bar{A}}_i^T\cdot\bar{\bar{A}}_j}_{\bar{\bar{A}}_{ij}})$$

$$\boxed{\mathcal{E}_H' = \frac{H^2}{2I_H} + \mathcal{U}}$$

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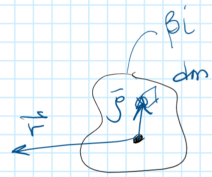
$$\mathcal{E}_H' = \frac{H^2}{2I_H} + \mathcal{U} \qquad \mathcal{U} = \frac{1}{2}\sum_{i=1}^{N}\sum_{j=1}^{N}\mathcal{U}_{ij}$$

generalises point mass to mass distr.

Very difficult to find closed form formulae for mutual potential

Gravitional potential of body

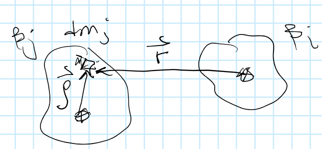
$$\mathcal{U}_i = -G\int_{\beta_i}\frac{dm}{|\vec{r} - \vec{\rho}|}$$



Known for
- sphere
- ellipsoid (const density)
- polyhedron (const density)
- spherical harmonics
- ellipsoidal "

mutual potential

$$\mathcal{U}_{ij} = \iint_{\beta_i\beta_j}\frac{dm_i\,dm_j}{|\vec{r}_{ij} - \bar{\bar{A}}_j\vec{\rho}_j - A_i\vec{\rho}_i|}$$



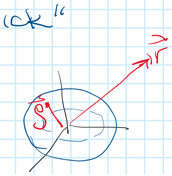
Know for
- sphere
- sphere + known potential
- self potential of ellipsoid
- series expansion
  - spherical harmonics
  - inertial integrals

In practice use series expansion

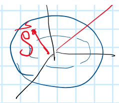
spherical body result "trick"

recall for sphere

$$\int_S\frac{dm}{|\vec{r} + \vec{\rho}|} = \frac{m}{|\vec{r}|}$$



$$\int_S \frac{dm}{|\vec{r}+\vec{\rho}|} = \frac{m}{|\vec{r}|}$$



for mutual potential

$$U_{ij} = -G \int_{\beta_i}\int_{\beta_j} \frac{dm_i\, dm_j}{\underbrace{|\vec{r}_{ij} + \bar{\bar{A}}_{ij}\vec{\rho}_j + \bar{\bar{A}}_i \vec{\rho}_i|}_{\vec{R}+\vec{\rho}}}$$

spherical object so can simplify

if β_i is a sphere

$$= -GM_i \int_{\beta_j} \frac{dm_j}{|\vec{r}_{ij} + \bar{\bar{A}}_{ij}\vec{\rho}_j|}$$

How do we use mutual potentials

Given mutual potential between 2 bodies we can generate relative forces & moments between them

$$\boxed{\vec{F}_{ij} = -\frac{\partial U_{ij}}{\partial \vec{r}_{ij}} \; ; \; \overset{\text{moment}}{\vec{M}_{ij}} = \frac{\partial U_{ij}}{\partial \theta_j}}$$

$\theta_j \approx$ infinitesimal rotations about each axis assuming a fixed $\bar{\bar{A}}_{ij}$ (orientation)

$U_{ij}(\vec{r}_{ij}, \bar{\bar{A}}_{ij})$

Symmetry

$$\vec{F}_{ij} = -\vec{F}_{ji}$$

$$\vec{M}_{ij} \neq \vec{M}_{ji} \Rightarrow \boxed{\vec{M}_{ij} + \vec{M}_{ji} + \vec{r}_{ij} \times \vec{F}_{ij} = \vec{0}}$$

Back to Energy, Ang Mom, Amended pot, etc.

KE:
$$T = \tfrac{1}{2} m_i r_i^2 + \tfrac{1}{2} \Omega_i \bar{\bar{I}}_i \cdot \Omega_i$$

$$T = \sum_{i=1}^{N} T_i = \frac{1}{2M} \sum_{1 \le i < j \le N} m_i m_j \dot{\vec{r}}_{ij}^{\,2} + \frac{1}{2}\sum_{i=1}^{N} \Omega_i \bar{\bar{I}} \cdot \Omega_i$$

Ang Mom:
$$\vec{H}_i = m_i \vec{r}_i \times \dot{\vec{r}}_i + \bar{\bar{A}}_i \cdot \bar{\bar{I}}_i \cdot \vec{\Omega}_i$$

Assume $\vec{R}_N = 0$ (barycenter)

$$\vec{H} = \sum_{i=1}^{N} \vec{H}_i = \frac{1}{M}\sum_{1 \le i < j \le N} m_i m_j \vec{r}_{ij} \times \dot{\vec{r}} + \sum_{i=1}^{N} \bar{\bar{A}}_i \cdot \bar{\bar{I}}_i \cdot \vec{\Omega}_i$$

Sundman's Inequality

$$\vec{H} = H \hat{H}$$

$$H^2 \le 2 T I_H \le 2 T I_p$$

Results hold independent of mass distributions
Hold for full NBP

$T = E - U$ $\quad$ U = summation over self & mutual potentials

$$H^2 \le 2(E-U) I_H \le 2(E-U) I_p$$

find the amended potentials

$$\left.\begin{aligned} \mathcal{E}_H &= \frac{H^2}{2I_H} + U \le E \\ \mathcal{E}_p &= \frac{H^2}{2I_p} + U \le E \end{aligned}\right\} \Rightarrow \mathcal{E}_p \le \boxed{\mathcal{E}_H \le E}$$

weaker bound on energy

"sharper"

$E = T_r + \mathcal{E}_p$

T_r = KE_{sys} relative to frame rotating instantaneously w/ $\vec{\Omega} = \frac{1}{I_H}\vec{H}$

$I_H(\vec{r}_{ij}, \bar{\bar{A}}_{ij})$ (not constant)

$T_r \equiv$ all KE in system w/ 'zero' ang momentum contribution

$\vec{\Omega} = \frac{1}{I_H}\vec{H}$

$T_r \equiv$ all KE in system w/ 'zero' ang. momentum contribution
i.e. radial velocities such as in the 2BP

$I_H(\vec{r}_{ij}, \bar{\bar{A}}_{ij})$ (not constant)

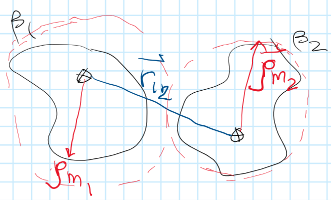
H is constant

$\mathcal{E}_H$ accounts for the KE induced by H^2

$\mathcal{E}_H$ is general, can be used to derive EOM of FNBP, can be used to find all relative equil of the FNBP as a function of H^2, can determine stability of any rel. equil. in the FNBP

Key Result in F2BP

any 2 mass distributions have minimum mass between them (assuming rigid bodies)



distance function

$d_{12} = |\vec{r}_{12}|$, need to add attitude

$d_{12}(\vec{r}_{12}, \bar{\bar{A}}_{12})$

$$\min d_{12}(\bar{\bar{A}}) = \boxed{d_{12}^{min} \leq \rho_{m_1} + \rho_{m_2}}$$

Thus if $r_{12} > \rho_{m_1} + \rho_{m_2} \Rightarrow$ no touching

Implies that there is a min PE between any 2 bodies,

$$\boxed{\mathcal{U}_{12} \geq \min_{\bar{\bar{A}}_{12}} \mathcal{U}_{12}(\vec{r}_{12}, \bar{\bar{A}}_{12}) = \mathcal{U}_{min}}$$

The existance of a min PE between 2 bodies changes functional properties of $\mathcal{E}_H$ & introduces new relative equil & new stable states

$\mathcal{E}_H = \frac{H^2}{2I_H} + \mathcal{U}$

for pt mass can have

$r_{12} \to 0 \Rightarrow \mathcal{U} \to -\infty$

$\mathcal{U}$ is unbounded from below

$r_{12} \to 0 \Rightarrow I_H \to 0 \Rightarrow \frac{H^2}{2I_H} \to \infty$ $\quad \mathcal{E}_H$ is undefined